PETW Phase Progress Report

PETW: A Multi-Agent Human-in-the-Loop Framework for Mechanism-Aware Thunderstorm Wind Engineering and Design Translation

Formal English phase report

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Executive Summary

PETW has expanded from a single research task into a multi-agent human-in-the-loop system for thunderstorm wind engineering research. Its primary scientific chain covers event classification, meteorological simulation, wind-field characterization, extreme-value analysis, engineering-scale transformation, structural response, and design-load translation. Zotero, LitWiki, Manuscript, MATLAB, and ComputeOps form the support layer. PI scientific judgment, operational authorization, and claim approval remain the final control layer.

As of August 10, 2026, the main phase achievements fall into three areas. First, the complete Step 0 through Step 5 scientific pipeline, bot responsibilities, data-packet interfaces, and inter-agent invocation mechanism have been established. Second, literature governance, visible inter-agent calls, task queues, HPC execution governance, deterministic document production, and scientific claim boundaries have been converted into an executable engineering system. Third, Events, CM1, WindTunnel, CFD, Structural, and DesignLoad have produced substantive research artifacts, including observational event packets, ensemble evidence from 40 Moore-style CM1 cases, wind-field and structural reviews, a UF wind-tunnel test matrix, CM1-to-CFD interface packages, a structural-response knowledge package, and GFF/G-GFF methodological prototypes.

PETW cannot yet claim a design wind speed for a physically validated event, `Vhat_3s`, return-period wind speed, structural-response correction factor, design-load expression, or code recommendation. The current system has primarily completed the research architecture, evidence governance, data interfaces, bounded simulations, and methodological validation. The principal gaps in the scientific chain remain event-mechanism confirmation, three-dimensional CM1 physical validation, engineering wind-speed standardization, common-event wind-field and loading evidence, an EVA-ready sample, structural-response calibration, and final design translation.

1 Recommended Title and Project Positioning

PI proposed title, `EnhancingDesignDownburstWindLoadingbyaMulti-agenthuman-in-the-loopframework`, accurately states the ultimate project objective. However, "Enhancing Design Downburst Wind Loading" can imply that improved design-load results have already been obtained. The following title is better aligned with the present phase:

> **PETW: A Multi-Agent Human-in-the-Loop Framework for Mechanism-Aware Thunderstorm Wind Engineering and Design Translation**

This title retains downburst research as a central direction while allowing PETW to treat gust fronts, propagating convective outflows, downburst-generated outflows, and mixed mechanisms explicitly. After the evidence chains for `Vhat_3s`, EVA, structural response, and design loading have been closed, PI original title can be used as a result-oriented manuscript title or as the final project title.

PETW has two linked objectives. The scientific objective is to establish an evidence chain from thunderstorm-wind mechanism identification to engineering design translation. The methodological objective is to determine how a multi-agent system can perform task decomposition, evidence retrieval, simulation preparation, execution governance, data processing, independent review, and claim control in complex engineering research while preserving human scientific responsibility.

2 Overall Scientific Architecture

The current primary scientific chain is:

```
Step 0 Event classification and reconstruction
    v Event Reconstruction Packet
Step 1 Mechanism-aware CM1/WRF meteorological simulation
    v Simulation Evidence Packet
Step 2 Wind-field characterization and resolution-adequacy assessment
    v Wind Field Feature Packet
Step 3 Mechanism-aware EVA and Vhat_3s standardization
    v EVA Return-Level and Uncertainty Packet
Step 4 Engineering-scale transformation and structural-response evidence
    |- Step 4a WindTunnel
    |- Step 4b CFD
    |- Step 4c Structural Response
    v WindTunnel, CFD Load, and Structural Response Packets
Step 5 DesignLoad synthesis and code interface
    v Candidate Design Translation and Claim Packet
PI      scientific review, claim approval, and external release
```

This chain is not a simple linear automation sequence. WindTunnel and CFD contain two complementary evidence paths:

1. A controlled accelerated-flow path investigates transient inflow, pressure, separation, reattachment, vortex shedding, and wake memory while producing validation evidence for CFD.
2. A meteorological thunderstorm-wind downscaling path transforms CM1/WRF or Step 2 wind fields into building-scale CFD boundary conditions, pressures, forces, and structural-response inputs.

The two paths share data formats, QA procedures, and pressure and structural interfaces, but they cannot substitute for one another. Accelerated-flow wind-tunnel evidence does not directly establish a real downburst wind field. Meteorological downscaling results cannot be translated directly into design loads without engineering-scale validation.

3 Human-in-the-Loop Control Layer

3.1 PI Role

PI retains the following authorities and responsibilities:

- Define research priorities and scientific questions.
- Determine whether event mechanisms, model configurations, and structural-response quantities are scientifically meaningful.
- Approve remote writes, HPC submissions, long-running tasks, facility operations, and high-impact data processing.
- Approve formal event identities, Path C results, `Vhat_3s`, EVA conclusions, structural-response factors, design-load expressions, and code language.
- Decide which results enter formal reports, manuscripts, design recommendations, or external releases.

PETW currently uses risk-tiered HITL rather than claim-by-claim approval. Independently

reviewed literature claims with strict source boundaries and preserved limitations may support internal synthesis and drafting. Source conflicts, cross-source extrapolations, formal validation, design decisions, code recommendations, and policy exceptions remain PI decisions.

3.2 Coordinator's Role

[path] converts PI objectives into bounded tasks, assigns a unique owner, selects the appropriate bot and route, checks input evidence, validates returned artifacts, maintains RCC state and handoffs, and prevents the following category errors:

- A plan is not an initiated task.
- Authorization is not execution.
- An input package is not a simulation result.
- An engineering test pass is not physical validation.
- HPC job completion is not an established scientific claim.
- A literature retrieval hit is not a complete reading.
- A candidate design expression is not design guidance.

3.3 Specialist Bot Roles

Specialist bots produce evidence, scripts, data packets, QA records, limitation statements, and PI decision prompts. They do not own final scientific interpretation and cannot promote a claim merely because workflow tests passed.

4 Data and Knowledge Architecture

PETW uses a layered source-of-truth model.

4.1 RCC Coordination Layer

Path: [path]

This layer stores compact project state, task packets, Agent Updates, handoffs, decisions, artifact paths, claim boundaries, and bot workspaces. Long logs, raw simulation outputs, and large datasets should not be accumulated directly in the coordination layer.

4.2 Zotero Literature Authority Layer

[path] is the authority for Zotero records, PDFs, and DOI identity. PETW bots do not receive unrestricted Zotero write access by default. `PETW-ZOTERO-AUTO-001` allows authorized sources to perform low-risk PETW literature maintenance, including deduplication checks, entry creation, completion of empty metadata fields, assignment to PETW collections, and attachment of legally accessible open PDFs. Deletion, merging, bulk reorganization, conflict overwrites, and acquisition of restricted PDFs still require separate PI approval.

4.3 LitWiki Semantic Knowledge Layer

Path: [path]

LitWiki converts Zotero literature into paper cards, source packets, claim records, conceptual relations, page-level source anchors, and searchable SQLite/FTS/Graph-RAG backends. In the most recent complete governance report, the governed active bundle contained 368 records, including 341 direct records, 27 canonical pointers, 334 verified direct records, and 7 source-specific blocked records. Governance CI was PASS with 0 findings, and 10 active retrieval entrypoints passed validation.

4.4 Scientific and Computational Workspaces

Events, CM1, WRF, Windfield, EVA, CFD, Structural, and DesignLoad each have dedicated workspaces. Raw simulations, wind-tunnel data, CFD cases, structural calculations, and scripts remain in the corresponding workspaces. RCC stores only compact summaries and paths.

4.5 Transaction and Visible-Invocation Layer

Inter-agent calls are stored under:

[path]

This layer contains outgoing payloads, incoming payloads, call states, the ledger, source outboxes, target inboxes, artifacts, submission provenance, and completion records. It makes bot calls traceable, idempotent, and reviewable, and it returns progress to the originating Feishu thread.

5 Interfaces and Calls Among Bots

5.1 Scientific Data-Packet Interfaces

The principal scientific interfaces currently include:

- `EventReconstructionPacket`: event window, station geometry, observed wind speed and direction, environmental changes, radar, satellite, and reanalysis evidence, candidate mechanisms, confidence, missing evidence, and downstream usability.
- `SimulationEvidencePacket`: input environment, triggering method, grid, physical parameters, run identity, raw outputs, failure evidence, spatiotemporal sampling, and the permitted range of scientific interpretation.
- `WindFieldFeaturePacket`: vertical profiles, horizontal footprint, near-surface turbulence, gust ramp, moving-outflow geometry, resolution state, and permitted downstream uses.
- `EVAReturn-LevelandUncertaintyPacket`: sample-inclusion rules, mechanism-label uncertainty, `Vhat_3s` conversion, peak under-sampling, mixed-climate structure, return-period sensitivity, and not-for-use warnings.
- `WindTunnelValidationPacket`: models, acceleration process, synchronized velocity, pressure, PIV, repeatability, uncertainty, and CFD comparison quantities.
- `CFDCaseManifest`: inflow source, mesh, boundary conditions, solver, time step, output frequency, validation source, pressure and force outputs, structural interface, and limitations.
- `CFDtoStructuralLoadPacket`: surface-pressure time histories or fields, integrated forces and moments, pressure-coefficient definition, mesh and time-step evidence, spatial correlation, and load-mapping assumptions.
- `StructuralResponsePacket`: structural prototype, response quantities, dynamic parameters, load source, response method, uncertainty, code baseline, and candidate `eta_{k,a,r}` status.

- `Step5DesignLoadHandoffPacket`: upstream evidence completeness, candidate expression, double-counting checks, applicability range, not-for-use status, and PI decision points.

5.2 Runtime Inter-Agent Invocation

The current general invocation chain is:

```
Source bot or Coordinator
-> Create a strict JSON IAC payload
-> petw_iac_post.py
-> HMAC-SHA256 local webhook
-> petw_iac_router.py
-> Route-registry and permission-ceiling checks
-> IAC ledger, source outbox, and target inbox
-> Kanban target-profile worker
-> Structured PROGRESS
-> Route-specific artifact
-> Artifact-hash and schema validation
-> Completion record
-> petw_iac_progress_relay.py
-> Originating Feishu thread or explicit callback surface
```

The local webhook is:

`http://127.0.0.1:8644/webhooks/petw-iac-router`

A live check on August 10, 2026 returned `{"status":"ok","platform":"webhook"}` from `/health`. The core `scripts.tests.test_petw_iac` test suite passed 31/31 tests.

The current `petw_iac_routes.json` uses `petw-iac-routes-v2` and contains 19 concrete routes, 9 aliases, and 17 allowed source profiles. Principal routes include:

- `petw.events.task`
- `petw.cm1.task`
- `petw.wrf.task`
- `petw.windfield.task`
- `petw.eva.task`
- `petw.cfd.task`
- `petw.structural.task`
- `petw.designload.task`
- `petw.litwiki.task`
- `petw.zotero.task`
- `petw.manuscript.task`
- `petw.figures.task`
- `petw.matlab.task`
- `petw.computeops.task`
- `petw.coordinator.task`
- `petw.remote.task`
- `petw.knowledge.task`

`petw.standardization.task` redirects to `petw.designload.task`.

5.3 Invocation Safety and Idempotency

The default constraints are:

```
remote_write_allowed = false
long_job_allowed = false
claim_approval_allowed = false
destructive_ops_allowed = false
max_depth = 1
```

Every payload must contain a unique `idempotency_key`. The router does not allow a request to raise a route's permission ceiling. The CM1 and CFD TACC routes contain separately locked conditions that restrict the source profile, TACC Stampede3 platform, `TG-CIV260040` account, and scientific domain. Deletion, cross-agent cancellation, scientific claim approval, and external submission remain prohibited or subject to separate approval.

5.4 Visible Progress Interface

Workers use a strictly structured progress format:

```
[PROGRESS] {"schema":"petw-iac-progress-v1","iac_id":"IAC-...","milestone_id":"...","
    summary":"..."}
```

The relay returns `ACCEPTED`, `RUNNING`, material `PROGRESS`, `PARTIAL`, `DONE`, or `BLOCKED` states to the originating Feishu thread. `HTTP202` means only that a task was accepted. It does not mean that the scientific work was completed. Final completion requires a route-specific artifact, matching `iac_id` and `task_id`, SHA-256, time boundaries, and a passing artifact validator.

6 Current Progress of Each Bot

6.1 Coordinator, [path]

The gateway is currently running. Coordinator has established the Step 0 through Step 5 division of responsibility, RCC workspaces, handoff and Agent Update systems, IAC router, HMAC webhook, Kanban worker, artifact validator, progress relay, Feishu thread return path, route-specific permissions, and the Zotero standing policy. Router health is currently normal, and the core tests passed 31/31.

Coordinator maturity is concentrated in research governance and invocation orchestration. It can coordinate Events, CM1, CFD, DesignLoad, LitWiki, Literature, ComputeOps, WRF, Manuscript, and other bots through auditable transactions. This achievement is research infrastructure, not scientific validation of downburst design loading.

6.2 Events, [path]

The gateway is currently running. Events has established layered labels, evidence confidence, detectability, mechanism uncertainty, and downstream-use boundaries. During M0/M1, a source table containing 879 rows and 19 columns was converted completely into 879 provisional `OBS` records, 879 evidence records, and 879 narrative records, with zero mismatches across the 19 original fields. This layer explicitly did not create `EVT`, mechanism truth, Path C, EVA-ready, or CM1-valid states.

M3 completed bounded evidence recovery for OKEM, BESSIE, and ERICK, covering 35 files and 873,948 bytes, followed by 32/32 verification. BESSIE then underwent observational reconstruction and a conditional CM1 case investigation. PI ultimately stopped further BESSIE package and CM1 progression and retained the case as an observational convective-outflow control.

The formal BESSIE state is that the observed `WMAX=34.9m/s`, convective association, and surface-outflow transition are established, while a propagating convective feature is supported. A downburst-specific reconstruction is not established. `VALID_DOWNBURST_CANDIDATE_LEVEL1` is not supported under the frozen gate. Path C is not promoted. There is no validated 10 m wind, 3 s gust, `Vhat_3s`, design guidance, or formal `EVT` identity.

6.3 CM1, [path]

The gateway is currently running. CM1 has completed a technical chain from compilation, micro smoke testing, and Thompson moist smoke testing through scale tests and large-case execution governance. Versioned private builds of CM1 r20.1 and r20.3 are available on TACC, and the execution, output, and NetCDF paths have been verified.

The Moore path currently contains 40 reconstructed cases. Of these, 24/40 satisfy the PETW Level-1 near-surface divergent-outflow candidate gate, 15/40 satisfy the `sws_max>=26m/s` strength proxy, and the clean denominator is 12/35. The two conditions are not nested. Formal three-dimensional downburst validation is 0/40, and `Vhat_3s` is 0/40. At least 35 of the 40 cases belong to the 97-member set in the Moore study, but the exact membership remains between 35 and 40 because the final 102-to-97 mapping has not been recovered.

The Opus 5 conclusion for the current ensemble is `ROUTE_A_ENSEMBLE_COMPARISON_PLUS_BOUNDED_LEVEL2_SHORTLIST_NO_NEW_PRODUCTION`. Four direct CFD-source candidates and one conditional candidate have been identified, but they are not formal three-dimensional validations, engineering-wind inputs, or `Vhat_3s` results.

BESSIE Job `3388245` is retained as a negative model-internal diagnostic. The run completed, but the frozen near-surface gate was not satisfied, and executable identity and complete source identity remain incomplete. It cannot be treated as a reconstruction of the observed event.

6.4 WRF, [path]

WRF is a PETW-controlled meteorological modeling support lane, and its gateway is currently running. This lane has established the WRF/WPS TACC toolchain, completed a real-data functional test, migrated Schumacher inputs, and governed a bounded extension job. The last explicit status artifact recorded Job `3371142` as `PENDING(Priority)` on August 4, 2026, configured for Stampede3 `skx`, 2 nodes, 64 tasks, and 16 h, with a registered ComputeOps watcher. This is the state at the timestamp of that evidence. It must not be described as still pending or as completed without a fresh scheduler readback.

WRF currently provides mechanism context, environmental reconstruction, and numerical-weather-modeling capability. No validated WRF wind field is yet available for entry into the PETW design-load chain.

6.5 Windfield, [path]

The gateway is currently running. Windfield has completed the Step 2 review and interface framework. The REV5 deliverable is a 53-page Chinese PDF with 116/116 admitted sources

cited, 15 original TikZ figures, 13 local tables, identical hashes from deterministic repeat builds, and 53/53 pages passing visual QA. All scientific states remain `not_approved` and `not_applied`.

Windfield has provided DesignLoad with point EPSD and Peng et al. vertical-coherence candidates. The current governed range supports only B2 sensitivity, with `f<=0.5Hz` and `delta_z<=100m`. B1 at `0.8Hz` and the full-height separation in B3 exceed the present evidence domain. The point EPSD, coherence, phase, and duration are not calibrated to the same event. Windfield has therefore established methodological and evidentiary boundaries, but it has not produced a complete `WindFieldFeaturePacket` that can directly support EVA, CFD, or design.

6.6 WindTunnel, [path]

The gateway is currently running. WindTunnel has progressed from conceptual planning to an experiment-preparation layer approaching facility handoff. The UF technical prefill package has been completed. The planned test articles are the M11, M31, and M51 section models, with Cobra probes, Scanivalve pressure acquisition, and 300 Hz stereoscopic PIV. The draft UF Test Matrix contains 9 worksheets and 540 populated rows, including 207 Tier 1 steady collections, 63 endpoint lookup collections at 11 m/s, 126 Tier 2 grouped sequences, 138 conditional W_TE PIV sequences, and 6 other conditional PIV placeholders.

The accelerated-flow knowledge layer now includes a 105-page, data-driven review of 55 sources, supported by 45 CSV files, 1,642 rows of newly extracted data, 16 new cross-study figures, and 56 source or existing figures. Final verification was PASS, and the PDF hash was fixed.

The current state remains a technical prefill and PI review candidate. No agent has operational authorization for wind-tunnel hardware, fans, the Flow Field Modulator, PIV, Scanivalve, gantry, turntable, or DAQ controls. No new PETW wind-tunnel experimental data or formal CFD validation packet has been produced.

6.7 CFD, [path]

The gateway is currently running. CFD has established a two-track manifest-driven automation system covering controlled accelerated-flow CFD and meteorological wind-field downscaling CFD. Current capabilities include input contracts, case manifests, an OpenFOAM source case, mesh and boundary checks, CM1 rank assembly, destaggering, postprocessing, validation comparison, and evidence packets.

The D420 CM1-to-CFD path completed the G2 read-only TACC preflight. Verification passed for 144 files, approximately 1.539 GB of represented source bytes, an 800 x 800 x 84 global grid, a 6 x 8 rank topology, and shared interfaces. A velocity-only neutral one-frame contract completed review, but it was explicitly classified as format and interface verification rather than a physically representative CFD input.

D420 M1/M0 postprocessing Job `3391286` failed on August 10, 2026 at the first M0 `reconstructPar`. The root cause was the absence of processor-level `constant/polyMesh/points` in all 48 processor directories while the root-level mesh existed. This was an execution-input contract defect, not a CM1 physical-mapping error, OpenFOAM divergence, or M1/M0 scientific difference. The job produced no reconstruction, comparison, solver result, `Vhat_3s`, or Figure 5-style scientific output. The one authorized submission was consumed. Any repair and resubmission require a new package and new PI authorization.

CFD now has relatively mature automation and governance capabilities, but it still lacks a validated storm-to-building pressure or load result.

6.8 Structural, [path]

The gateway is currently running. Step 4c exact146 v9 is the current PI evidence package for governed activation and handoff. It contains 146/146 accepted records, 0 `correcti on_required`, 0 `pending_review`, and 0 blocked. The active non-final wiki publication updated 27/27 targets, Governance CI was PASS, and 153/153 tests passed.

PI approved Option A and Option B, covering publication of evidence status and use of literature-supported background and methods-governance claims in the phase report, manuscript planning, and HITL case-study descriptions. PI has not approved PETW mechanism conclusions, structural-response results, design guidance, `eta_k`, or code recommendations.

Structural also supplied DesignLoad with methodological inputs for B1 through B3, including structural geometry, frequency, damping, `rho_B=180kg/m3`, `phi1=z/H`, and candidate `m(z)`. These inputs passed unit and integration checks. However, the absence of source-bound aerodynamic generalized force or force EPSD prevents closure of displacement, base shear, overturning moment, ESWL, or GFF/G-GFF calibration.

6.9 DesignLoad, [path]

The gateway is currently running. DesignLoad has absorbed the former Standardization lane and established a mechanism-aware design-load and code-interface framework. The current governed knowledge layer contains a 412-record registry and a 94-document SQLite/FTS collection. PI reviewed the bounded use of the DL5 claim packet but did not approve any design-load expression, numerical coefficient, reliability factor, conservative default, or code-replacement recommendation.

The GFF/G-GFF prototype implements the I1 through I4 decomposition, the ASCE 7-22 baseline, the Kwon-Kareem method, and a Windfield EPSD candidate. B2 full time-frequency sensitivity was completed within the governed range, with tests, benchmarks, RK4 integration, and independent review passing. The B2 normalized response sensitivity, assumption-driven dimensional demonstration, and C0 calibration screening are methodological evidence.

The most important current boundary is the absence of common-event point EPSD, vertical coherence, phase, duration mapping, surface pressure, generalized force, and a structural-response target. B1 and B3 remain blocked by evidence-range limitations. DesignLoad has reached mathematical and software closure C0, but not common-event evidence closure C1, physical-model closure C2, or design-governance closure C3.

6.10 LitWiki, [path]

The gateway is currently running. LitWiki has evolved from a first-pass file library into a governed semantic knowledge layer. The latest complete governance report recorded 368 active records, 334 verified direct records, 7 blocked direct records, 27 canonical pointers, 10 active retrieval entrypoints, SQLite integrity `ok`, and Governance CI with 0 findings.

LitWiki covers namespaces including Step 0, Step 1, CM1-to-CFD, CFD-M1, CFD-M3, CFD-M4, Step 4c, and Step 5. The system distinguishes PDF availability, text extraction, candidate reading, independent review, canonical reuse, blocked sources, and scientific approval. LitWiki maturity is relatively high, but verified literature does not automatically become PETW physical validation or design guidance.

6.11 Literature/Zotero, [path]

This profile is not PETW-specific, but its gateway is currently running and it is the authority for PETW literature identity, PDFs, DOI records, and Zotero writes. Events, LitWiki, and Coordinator can invoke its low-risk maintenance capabilities through `petw.zotero.task` and `PETW-ZOTERO-AUTO-001`. The completion path includes DOI deduplication checks, metadata completion, collection assignment, legally accessible open-PDF attachment, and a compact ACK. High-risk operations remain subject to PI approval.

6.12 Manuscript, [path]

The gateway is currently running. Manuscript has established a multistage, reviewable document-production pipeline containing source packets, model-use records, claim-safety review, LaTeX/Tectonic compilation, deterministic builds, citation audits, figure-rights checks, and visual QA.

Representative current deliverables include:

- Windfield Step 2 REV5, 53 pages with 116/116 citation coverage.
- Structural Step 4c exact146 review, 22 pages with mapping to 146/146 accepted sources.
- Accelerated-flow author-date review, 96 pages with 55 citation keys, 56 numbered figures, 18 comparison tables, and 18/18 QA gates passing.
- Chinese and English PETW agentic HITL proposal baselines and a centralized automated writing workflow.

These deliverables establish document automation and evidence-integration capability. They do not indicate that the candidate scientific claims in the documents have received final approval.

6.13 MATLAB, [path]

The gateway is currently running. The MATLAB bot has completed Lark DM configuration, workspace setup, and scripting-support preparation. It can receive `.m` scripting, time-series processing, turbulence-statistics, and data-interface tasks from Windfield, EVA, Structural, DesignLoad, and ComputeOps. Current durable evidence primarily establishes runtime capability and interface readiness rather than an independent central scientific result.

6.14 ComputeOps, [path]

The gateway is currently running. ComputeOps has established connection, ControlMaster, account, partition, storage, transfer, queue, Slurm, and evidence runbooks for Notre Dame CRC, HTCondor, and TACC Stampede3. Controlled platform capabilities have been verified for WRF/WPS, CM1, OpenFOAM, and Fluent.

ComputeOps owns platform and execution evidence, not scientific interpretation. Remote writes, submissions, cancellations, long-running tasks, and large transfers still require specific authorization. The dedicated CM1 and CFD TACC routes lock the allowed platform, account, source profile, domain, and prohibited operations into the route registry.

7 Phase Completion Assessment

7.1 Components with Stable Operational Capability

- Step 0 through Step 5 research responsibilities and data-packet interfaces.
- Coordinator, RCC, handoff, Agent Update, and task-owner mechanisms.
- A visible, idempotent, and verifiable IAC router and Feishu progress relay.
- Governed Zotero and LitWiki literature knowledge layers.
- CM1, WRF, OpenFOAM, and Fluent HPC toolchains and execution governance.
- Manuscript multi-model collaboration, deterministic LaTeX/PDF builds, citation QA, and figure QA.
- WindTunnel UF test planning, technical prefill, and large-scale literature data extraction.
- DesignLoad methodological prototypes and not-for-design boundaries.

7.2 Components with Scientific or Data Evidence but Incomplete Closure

- Evidence recovery for 879 provisional observation anchors and three event anchors.
- The BESSIE observational convective-outflow control case.
- Forty Moore-style CM1 reconstructed cases and 24 Level-1 candidates.
- A bounded CM1-to-CFD D420 velocity interface and source-root evidence.
- Bounded methodological evidence for wind-field EPSD and coherence.
- Governed Structural exact146 methods and background evidence.
- The DesignLoad B2 methodological demonstration and C0 screening.

7.3 Scientific Loops Not Yet Closed

- A PI formal event catalog and mechanism labels.
- A validated three-dimensional downburst simulation.
- Observation-equivalent 10 m wind, 3 s gust, and `Vhat_3s`.
- A mechanism-conditioned sample suitable for EVA.
- Mechanism-conditioned return levels or design wind speeds.
- A validated storm-to-building pressure or load history.
- PETW evidence-driven structural response.
- Calibrated `eta_{k,a,r}`, C-GFF, ESWL, or design-load expressions.
- ASCE 7 or Chinese code recommendations.

8 Current Major Bottlenecks

First, the event layer still lacks a unified closure rule for promoting provisional observations to formal independent events and mechanism labels. BESSIE demonstrates that a strong convective-outflow observation can be established while a downburst-specific mechanism remains unresolved.

Second, CM1 has substantial ensemble and execution capability, but formal three-dimensional physical validation, observation-comparable quantities, and `Vhat_3s` remain absent. Internal `sws`, `sws2`, or lowest-model-level wind cannot directly substitute for an engineering 3 s gust.

Third, Windfield has a comprehensive review and candidate models but lacks a formal feature

packet for one common event with unified units, spatiotemporal sampling, and explicit downstream uses.

Fourth, EVA remains at the planning stage. Return levels should not be calculated without an upstream event sample, `What_3s`, and an uncertainty packet.

Fifth, the D420 CFD chain remains blocked by execution inputs and the decomposed-mesh contract. The current failure produced no scientific output, and additional engineering hardening cannot substitute for actual CFD data.

Sixth, Structural and DesignLoad have relatively complete methodological systems but lack common-event wind-field, pressure, generalized-force, and response targets. They currently support only methodological demonstration and C0 software closure.

Seventh, WindTunnel is approaching facility handoff, but the actual experiment, facility authorization, channel mapping, PIV conditions, and execution matrix have not been frozen.

9 Recommended Next Phase

9.1 Priority Task A: Moore 40-Case Ensemble Comparison and Level-2 Shortlist Closure

Scientific question: Can the existing 40 CM1 cases support interpretable mechanism, strength, boundary, and CFD-source classifications without new production runs. **Existing evidence:** 40 reconstructed cases, 24 Level-1 candidates, 15 strength-proxy cases, 4 direct Level-2 candidates, and 1 conditional candidate. **New evidence required:** A unified ensemble comparison, repair of figure-blocked cases, boundary-clearance records, and a source-suitability packet for each Level-2 candidate. **Existing method:** Use the frozen `PETW_MOORE_UNIFORM_AGL_V1` data, the existing Level-1 gate, and the Opus 5 Route A decision. **Minimum checks:** Case identity, 40 unique IDs, exclusion of run 798, denominator definitions, boundary distance, footprint, output cadence, and claim labels. **Actual blocking conditions:** Unreadable local outputs, unresolved case identity, missing candidate-dependent fields, or a comparison that would alter the frozen gate. **Completion condition:** All 40 cases have a unified comparison row, and every Level-2 shortlist candidate has an explicit retention or exclusion rationale. **Stopping condition:** Complete the ensemble comparison using existing outputs without adding a CM1 production batch. **Claim boundary:** Produce PETW internal candidates only, without three-dimensional validation, engineering-wind inputs, or `What_3s`. PI **decision after completion:** Decide whether one Level-2 candidate may enter a bounded CFD-source packet.

9.2 Priority Task B: Step 2 Single-Case Wind Field Feature Packet v1

Scientific question: Which wind-field quantities from one selected CM1 candidate can be extracted credibly at the present resolution and cadence, and which may enter CFD, EVA, or Structural. **Existing evidence:** Windfield REV5, outputs from the 40 CM1 cases, D420 interface experience, and the existing EPSD and coherence method boundaries. **New evidence required:** A single-case packet containing profiles, footprint, ramp, time history, resolution status, sampling operator, and downstream-use flags. **Existing method:** Apply the Windfield M1/M2/E scales and resolution-adequacy gate. **Minimum checks:** Coordinates, units, reference height, time window, cadence, destaggering, boundary effects, finite values, observational equivalence, and output identity. **Actual blocking conditions:** A target quantity below model resolution, an undefined sampling operator, incomplete source-case identity, or a quantity that can only serve as a proxy. **Completion condition:** Every feature is classified as resolved, parameterized, proxy-only, or excluded, with permitted and prohibited downstream uses. **Stop-**

ping condition: Complete one packet without expanding into batch surrogate training. **Claim boundary:** Do not claim `Vhat_3s`, an engineering gust, or a design wind speed. PI **decision after completion:** Decide whether the packet enters a CFD pilot, an EVA schema test, or remains methods evidence only.

9.3 Priority Task C: Executable EVA Data-Contract Validation

Scientific question: Can the current upstream packets satisfy the minimum data contract for mechanism-aware EVA. **Existing evidence:** EVA planning artifacts, Events labels, CM1 outputs, the Windfield packet schema, and DesignLoad not-for-use rules. **New evidence required:** A small field-by-field validated EVA pilot table that may be empty or contain a small number of non-inferential records while explicitly representing missingness and inclusion status. **Existing method:** Use `EVA_Data_Contract_v1`, the `Vhat_3s` standardization specification, and the mechanism-classification uncertainty fields. **Minimum checks:** Event independence, mechanism status, peak definition, sampling operator, units, observational or simulation origin, uncertainty, inclusion flag, and forbidden-use flag. **Actual blocking conditions:** Unassessed event independence, absence of `Vhat_3s`, incomplete source identity, or treatment of a mechanism label as truth. **Completion condition:** The data contract passes field and logical validation and identifies which rows may enter future EVA and which can only be placeholders or sensitivity inputs. **Stopping condition:** Do not calculate return levels, fit GEV/POT models, or generate a design wind speed. **Claim boundary:** Establish only that the data contract is executable, not that a statistical conclusion is supported. PI **decision after completion:** Decide whether to expand the event sample or return to Step 0 or Step 2 to obtain additional evidence.

9.4 Priority Task D: UF WindTunnel Pre-Execution Review

Scientific question: Are the current UF form and 540-row matrix sufficient to freeze the first controlled accelerated-flow experimental stage. **Existing evidence:** UF technical prefill, test matrix, v5.8 plan, 55-source review, and identified TBC fields. **New evidence required:** A PI event structure, model order, pressure repeats, PIV activation plan, Cobra reference, turntable configuration, FFM configuration, and synchronization plan. **Existing method:** Use the current UF template and facility-owner review without creating a new facility-control adapter. **Minimum checks:** Channel map, trigger synchronization, speed range, models and angles, PIV FOV, repeatability rules, facility limitations, personnel authorization, and safety boundaries. **Actual blocking conditions:** UF has not confirmed the control interface, PIV and pressure synchronization, model installation, or safety rules. **Completion condition:** Produce a facility-approved test matrix and an explicit first-run subset. **Stopping condition:** Do not generate or send facility-control commands before approval by the UF operator and Peter. **Claim boundary:** Approval of the experimental plan does not mean that the experiment or CFD validation has been completed. PI **decision after completion:** Approve the first facility-execution subset or require modification.

10 Current PI Decisions Required

1. Use the claim-safe project title in this report or retain PI original wording as the final objective-oriented title.
2. Select the next scientific center among the Moore 40-case ensemble comparison, one Windfield feature packet, an EVA data-contract pilot, or the UF WindTunnel execution review.

3. Decide whether D420 Job `3391286` warrants a new package and a new submission approval. The current default recommendation is not to prioritize it before ensemble and Step 2 closure.
4. Decide whether this report should also be converted into a formal LaTeX/PDF version, a Feishu document, or a shorter version for an advisor or collaborator.

11 Concluding Remarks

PETW has established an operational multi-agent HITL research system. It now has a defined Step 0 through Step 5 scientific chain, specialist bot responsibilities, a layered knowledge base, strict data-packet interfaces, a visible IAC router, HPC governance, literature governance, and deterministic report-production capability. Events, CM1, WindTunnel, CFD, Structural, and DesignLoad have produced reviewable artifacts, and the system has demonstrated actual inter-bot calls and feedback loops.

The strongest current achievements are the research architecture, evidence governance, inter-agent collaboration, and bounded scientific data production. The most important remaining scientific task is to move from mechanism-uncertain events and model-internal quantities to a unified engineering wind-field packet, `Vhat_3s`, EVA, pressure and structural-response evidence, and ultimately design loading. The next phase should minimize new infrastructure and use existing methods and data to complete one small but scientifically complete loop.

Principal Evidence Paths

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